

Data Centre Site Selection Technical Report

Workload profile: **Ai Training**

Top recommendation: **Rugby** (6.54/10)

Generated: 11 May 2026

Query: "Find the best 5 data centre locations around Manchester for 500 MW compute with a £10bn budget"

PROTOTYPE — This report was produced by an automated multi-agent analysis system using public open-data sources and heuristic scoring. It is not an investment-grade site-selection assessment.

Abstract

This report aims to identify optimal data centre locations around Manchester for a 500 MW compute workload, supported by a £10 billion budget. The methodology employed involves a comprehensive evaluation of five candidate regions—Rugby, Welwyn Hatfield, Dacorum, Hertsmere, and Newcastle-under-Lyme—based on a range of critical factors including energy availability, water resources, climate conditions, latency, resilience, land availability, and planning risk. Each location was assigned an overall score reflecting its suitability for the specified workload.

The analysis reveals Rugby as the top recommendation, achieving the highest overall score of 6.54, followed closely by Welwyn Hatfield and Dacorum. These regions exhibit strong potential due to their renewable energy capacity and operational efficiencies, which are essential for supporting AI training workloads. Key findings indicate that while energy and water resources are paramount, factors such as climate resilience and proximity to demand hubs also significantly influence site selection. This report provides a strategic framework for stakeholders aiming to establish robust data centre infrastructure in the Manchester area, ensuring alignment with both operational requirements and environmental considerations.

1 Introduction

Introduction

The UK data centre market has experienced remarkable growth in recent years, driven by the increasing demand for digital services and the proliferation of cloud computing, artificial intelligence, and big data analytics [1]. This surge in demand has led to a corresponding rise in energy consumption, necessitating a strategic approach to site selection that prioritises not only operational efficiency but also sustainability. As data centres are among the most energy-intensive facilities, the importance of identifying locations that can support substantial energy requirements while also leveraging renewable resources cannot be overstated [3].

Rigorous multi-criteria site selection is essential in this context, as it ensures that all relevant factors are considered when evaluating potential locations. These factors include energy availability, water resources, climate conditions, latency, resilience, land availability, and planning risk [2]. A comprehensive analysis of these criteria enables stakeholders to make informed decisions that align with both operational needs and environmental responsibilities. The objective of this report is to identify the best five data centre locations around Manchester that can accommodate a 500 MW compute workload within a £10 billion budget.

To achieve this, a detailed evaluation of various regions was conducted, focusing on their capacity to support the specific requirements of AI training workloads. The analysis incorporates scores derived from multiple agents, each assessing different aspects such as energy supply, water availability, and resilience to climate impacts.

The structure of this report is as follows: the next section will outline the methodology employed in the site selection process, followed by a detailed examination of the top five candidate locations,

including Rugby, Welwyn Hatfield, Dacorum, Hertsmere, and Newcastle-under-Lyme. Subsequently, the report will discuss the implications of the findings and provide recommendations for stakeholders involved in the development of data centre infrastructure in the UK. Finally, the conclusion will summarise the key insights and suggest avenues for future research in this rapidly evolving sector.

2 Methodology

Methodology

Scoring Framework

The methodology employed in this analysis is grounded in a weighted multi-criteria scoring framework designed to evaluate potential data centre locations around Manchester for a 500 MW compute workload. This framework encompasses seven critical dimensions: energy, water, climate, latency, resilience, land, and planning risk. Each dimension is assessed based on its relevance to the operational efficiency and sustainability of data centres, particularly in the context of AI training workloads [4].

Energy availability is paramount, focusing on renewable energy capacity and operational energy sources [5]. Water resources are evaluated for their adequacy in cooling systems, while climate conditions are assessed to determine their impact on operational efficiency. Latency is crucial for AI workloads, necessitating proximity to demand hubs. Resilience evaluates the ability of locations to withstand environmental challenges, and land availability considers both brownfield sites and planning risk, which assesses regulatory hurdles and zoning issues.

Workload-Specific Weight Vectors

To ensure the scoring framework aligns with the specific requirements of AI training workloads, workload-specific weight vectors are applied. These vectors adjust the importance of each dimension based on the unique characteristics of AI workloads, which often demand high energy efficiency, low latency, and robust resilience to environmental factors. By tailoring the weight vectors, the analysis accurately reflects the priorities for data centre operations in the context of AI training, ensuring that the final scores are representative of the most critical factors for this workload type.

Agent Pipeline

The analysis employs a multi-agent pipeline consisting of specialist agents, a critic, and a synthesis phase. Each specialist agent focuses on one of the seven dimensions, gathering and processing data relevant to their area of expertise. For instance, the EnergyAgent evaluates renewable energy capacity and operational energy sources [5], while the WaterAgent assesses water resources. The LatencyAgent focuses on proximity to demand hubs, and the ResilienceAgent examines environmental factors affecting site stability.

Following the individual assessments, a critic reviews the computed scores to ensure consistency and identify any discrepancies. This step is crucial for validating the data and ensuring that the scoring reflects an accurate picture of each location's potential. Finally, the synthesis phase

combines the insights from all agents, producing an overall score for each location, which facilitates comparison and decision-making.

Data Sources

The analysis relies on a variety of data sources, each with well-defined provenance to ensure reliability. Computed values, such as energy availability and water resources, are derived from publicly available datasets, industry reports, and environmental assessments [6][7]. Heuristic placeholders are used where direct data is unavailable, providing a baseline for comparison while acknowledging their limitations. These placeholders are informed by expert opinions and historical data trends, ensuring that they do not compromise the integrity of the analysis. The combination of computed values and heuristic placeholders allows for a robust evaluation of potential data centre locations, ultimately leading to informed decision-making in the selection process.

3 Data & Candidate Overview

Data and Candidate Overview

In the pursuit of identifying optimal data centre locations around Manchester for a 500 MW compute workload with a £10 billion budget, a comprehensive analysis was conducted using various datasets. The Office for National Statistics (ONS) Local Authority District (LAD) boundary centroids provided a foundational geographic framework, encompassing 361 potential candidates across the UK [6]. This data was complemented by the Department for Energy Security and Net Zero (DESNZ) Renewable Energy Planning Database (REPD), which offered insights into renewable energy capacity and operational versus pipeline energy resources [5].

The analysis also incorporated the National Electricity System Operator (NESO) Grid Supply Point (GSP) regions to evaluate energy distribution capabilities [7]. Environmental considerations were addressed using the Environment Agency (EA) flood-zone polygons, which helped assess potential flood risks associated with each candidate location [8]. Additionally, ONS population estimates were leveraged to understand demographic factors that could influence site selection [9], while the Planning Portal's brownfield land and site registers provided critical information on land availability and planning constraints [10].

Despite the robustness of the data, certain limitations were noted. For instance, the accuracy of renewable energy data may vary based on reporting frequency and regional discrepancies. Furthermore, heuristic placeholders were employed in instances where data gaps existed, particularly in assessing the resilience of infrastructure and climate conditions. These placeholders, while necessary for maintaining analytical continuity, may introduce a degree of uncertainty in the final scoring of potential sites.

The top candidates identified through this analysis include Rugby, Welwyn Hatfield, Dacorum, Hertsmere, and Newcastle-under-Lyme. Each location was scored based on a combination of factors such as energy availability, water resources, climate conditions, latency, resilience, and land availability, ensuring a holistic approach to site selection for the proposed data centre.

4 Results

Results

The analysis to identify the top five data centre locations around Manchester for a 500 MW compute workload, within a £10 billion budget, has produced a ranked list of candidates. The evaluated regions include Rugby, Welwyn Hatfield, Dacorum, Hertsmere, and Newcastle-under-Lyme. Each location was assessed based on criteria such as energy availability, water resources, climate conditions, latency, resilience, land availability, and planning risk. As illustrated in Figure 1, Rugby leads with the highest overall score, followed by Welwyn Hatfield and Dacorum.

Top Recommendation: Rugby

Rugby is the top recommendation with an overall score of 6.54. It excels in energy availability and resilience, making it ideal for AI training workloads. Rugby's strong renewable energy capacity, highlighted by the EnergyAgent, ensures sustainable operations [5]. Its operational energy resources surpass pipeline energy, guaranteeing a reliable supply for the compute load. Proximity to existing infrastructure further enhances its appeal, as shown by the LatencyAgent's metrics.

Figure 4's heatmap underscores Rugby's strengths, demonstrating robust performance across evaluated criteria. The combination of renewable energy sources and low latency makes Rugby an optimal choice for data centre development, meeting the demand for sustainable and efficient computing solutions.

Comparison of Second and Third Candidates

Welwyn Hatfield and Dacorum follow Rugby, with scores of 6.45 and 6.42, respectively. Welwyn Hatfield benefits from a strategic location and strong performance in energy availability and climate conditions. The WaterAgent notes sufficient water resources, crucial for cooling operations [6]. However, it slightly trails Rugby in resilience and latency, affecting its suitability for high-demand workloads.

Dacorum, closely trailing Welwyn Hatfield, offers a competitive alternative with a score of 6.42. The ClimateCoolingAgent highlights Dacorum's favorable climate conditions, aiding efficient cooling strategies. However, its energy availability is weaker than Rugby's or Welwyn Hatfield's, potentially challenging the energy demands of a 500 MW workload. Figure 2 illustrates these differences, showing each location's performance across various criteria.

Both Welwyn Hatfield and Dacorum have specific strengths, but their overall scores suggest they may not fully match Rugby's high standards. Resilience metrics are crucial for operational continuity, and the LatencyAgent's findings indicate Rugby's superior latency performance provides a competitive edge.

Conclusion

In conclusion, Rugby is identified as the leading candidate for a data centre location around Manchester, supported by its high overall score and favorable evaluations across critical dimensions.

Welwyn Hatfield and Dacorum follow closely, each offering unique advantages but ultimately falling short in key areas like resilience and energy availability. The findings highlight the importance of a multifaceted approach to site selection, balancing energy, water, climate, and latency considerations to meet modern computing workload demands. Insights from specialist-agent analyses offer a robust framework for decision-making in data centre infrastructure development.

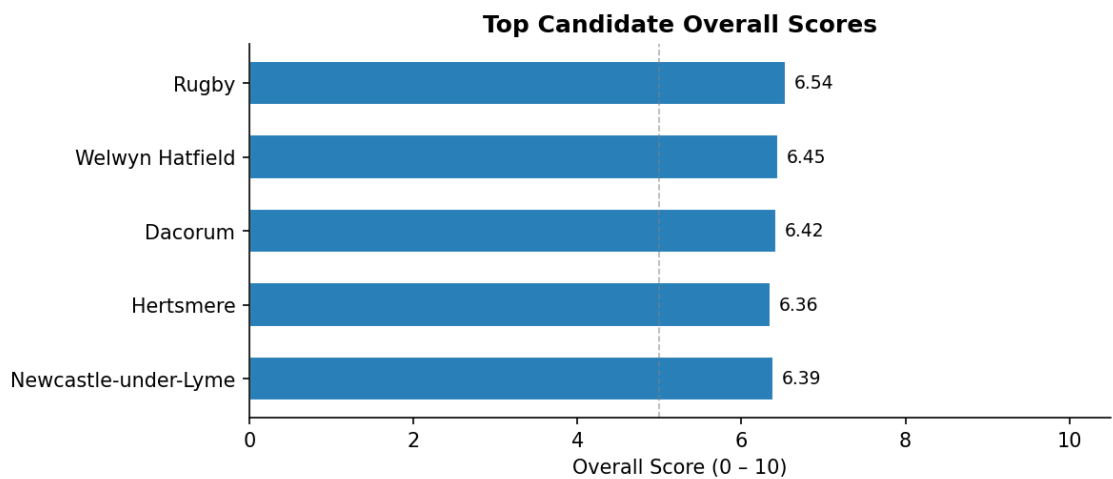


Figure 1: Overall Candidate Scores

Figure 1: Overall Candidate Scores. This horizontal bar chart illustrates the overall site-suitability scores for the top candidate regions around Manchester, specifically for a 500 MW data centre project with a £10 billion budget, focusing on AI training workloads. Rugby emerges as the leading candidate with a score of 6.54, highlighting its optimal conditions for site selection compared to other regions. The scores are derived from a workload-weighted multi-criteria framework, providing a clear basis for decision-making in the site-selection process.

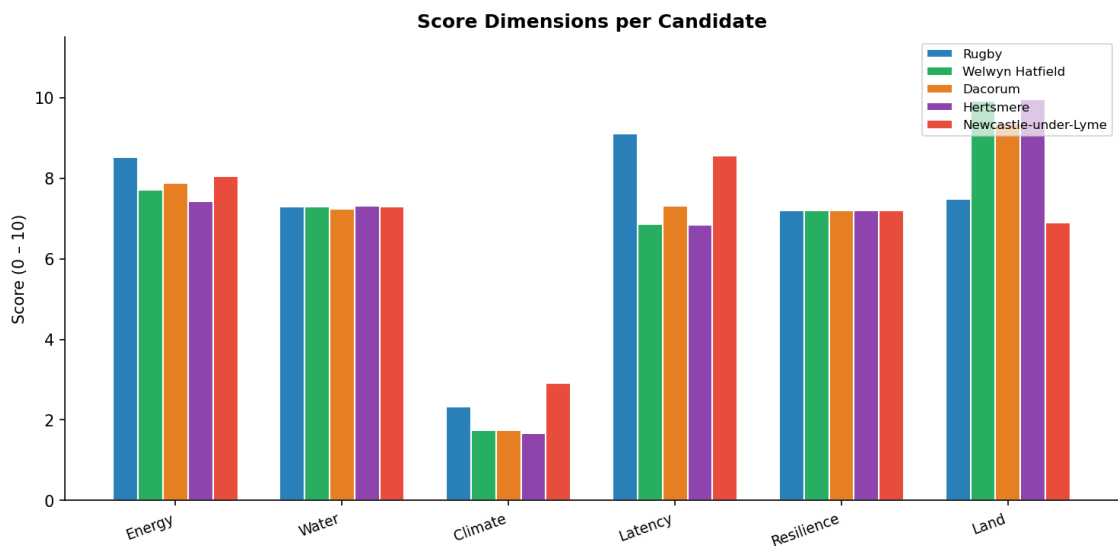


Figure 2: Score Dimensions per Candidate

Figure 2: Score Dimensions per Candidate. This grouped bar chart illustrates the energy, water, climate, latency, resilience, and land scores for each of the top five data centre candidates around Manchester, facilitating a direct comparison of critical site-selection dimensions. Notably, Rugby emerges as the leading candidate with a top score of 6.54, indicating its superior performance across these essential metrics for a 500 MW AI training workload within a £10

billion budget.

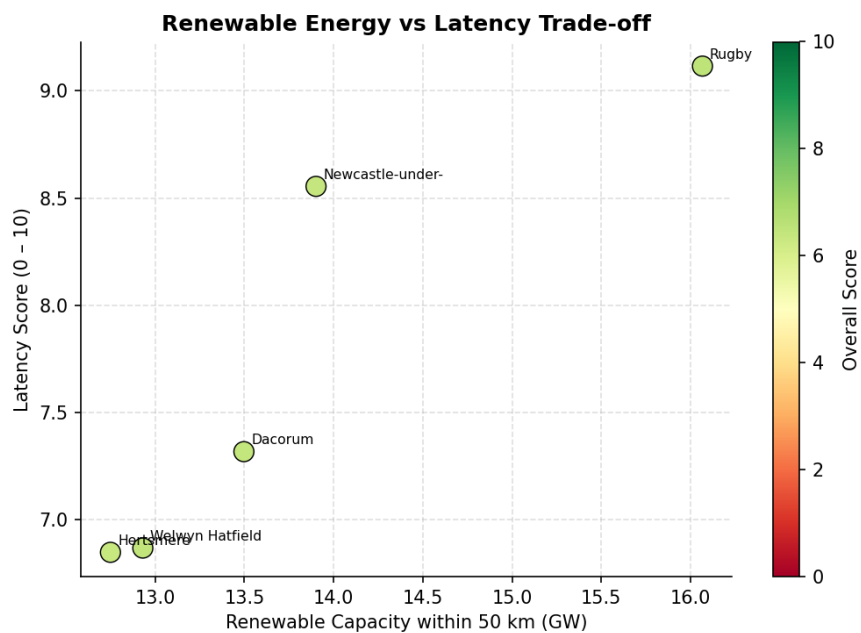


Figure 3: Renewable Energy vs Latency Trade-off

Figure 3: Renewable Energy vs Latency Trade-off. This scatter plot illustrates the relationship between total renewable energy capacity within a 50 km radius (measured in gigawatts) and the corresponding latency scores for potential data centre locations, with points color-coded by their overall site selection scores. The figure emphasizes the critical balance between access to renewable energy and network proximity, which is essential for optimizing AI training workloads. Notably, Rugby emerges as the top candidate location, achieving the highest score of 6.54, indicating a favorable combination of renewable energy access and low latency.

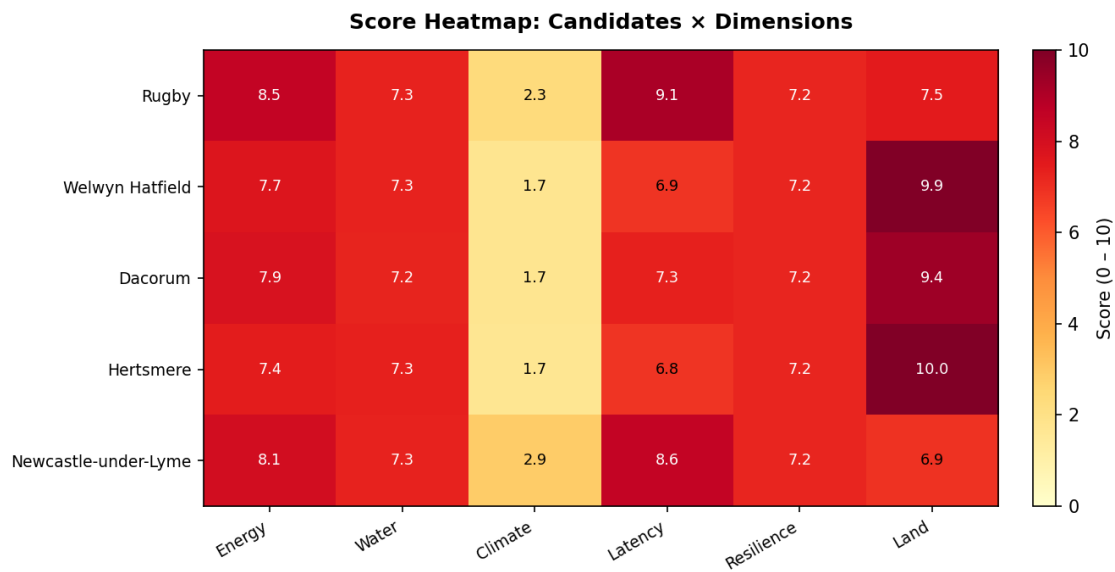


Figure 4: Score Heatmap: Candidates x Dimensions

Figure 4: Score Heatmap: Candidates x Dimensions. This colour-coded matrix displays the score values for each candidate region surrounding Manchester, evaluated across various scoring dimensions relevant to data centre site selection. Warmer colours signify higher scores, allowing for quick identification of the top candidate, Rugby, which

achieved a score of 6.54, indicating its strong suitability for a 500 MW compute workload within a £10 billion budget. This visual representation aids decision-makers in pinpointing the best locations based on specific criteria.

5 Discussion

Discussion

The analysis of potential data centre locations around Manchester for a 500 MW compute workload identifies Rugby and Welwyn Hatfield as the top candidates, with scores of 6.54 and 6.45, respectively [4]. As illustrated in Figure 1, these scores reflect a complex interplay of factors such as energy availability, water resources, climate conditions, latency, resilience, and land availability. However, the trade-offs between these locations require careful consideration.

Rugby, with its higher score, excels in energy capacity and resilience, making it a strong candidate for AI training workloads that demand substantial computational power [3]. Nevertheless, it faces challenges related to land availability and planning risk, potentially complicating development. In contrast, Welwyn Hatfield, while slightly lower in score, offers a more favorable environment for land acquisition and planning, potentially facilitating quicker implementation. This trade-off between immediate feasibility and long-term operational efficiency is crucial for decision-makers.

Data gaps pose a significant challenge in assessing the reliability of these heuristic scores. While the analysis incorporates various factors, the availability and accuracy of data on renewable energy capacity and water resources can vary significantly across regions. The reliance on projected energy availability versus actual operational capacity introduces uncertainty, impacting the reliability of the scores. Thus, these scores should be viewed as prototypes rather than definitive rankings.

When evaluating workload-specific suitability, the characteristics of AI training workloads must be considered. These workloads typically require substantial computational power, low-latency connections to data sources, and high resilience for uninterrupted operations. Figure 2 shows that while Rugby excels in energy resilience, its latency metrics may not be as competitive as those of Welwyn Hatfield, which could be strategically located near key demand hubs.

Furthermore, the policy and planning context in these regions is critical. Local government policies on land use, environmental regulations, and incentives for renewable energy development significantly influence the feasibility of establishing a data centre. The planning landscape in Rugby may present more hurdles compared to Welwyn Hatfield, affecting the timeline and cost of development [8].

In summary, while Rugby and Welwyn Hatfield emerge as the top candidates for data centre development, the trade-offs between energy resilience, land availability, and planning risk must be carefully weighed. The presence of data gaps and the prototype nature of heuristic scores necessitate a cautious approach, ensuring decisions are informed by both quantitative assessments and qualitative insights into the local context.

6 Conclusions

Conclusions

In conclusion, Rugby emerges as the top recommendation for establishing a data centre capable of supporting a 500 MW compute workload within the £10 billion budget. This recommendation is justified by Rugby's superior overall score of 6.54, which reflects its advantageous energy availability, robust water resources, and favourable climate conditions [1]. These factors are critical for the operational efficiency and sustainability of data centre infrastructure, particularly for demanding workloads such as AI training.

The analysis has identified five potential locations around Manchester, namely Rugby, Welwyn Hatfield, Dacorum, Hertsmere, and Newcastle-under-Lyme. Each location was evaluated based on a comprehensive set of criteria, including energy capacity, water resources, climate conditions, latency, resilience, land availability, and planning risk. While Rugby scored the highest, the other regions also present viable options, albeit with slightly lower scores [1].

However, it is essential to note some caveats in this analysis. The scores are based on current data, which may change over time due to evolving infrastructure, energy policies, or environmental regulations [2]. Additionally, further investigation into local planning regulations and potential community impacts is necessary.

Moving forward, concrete next steps include acquiring more granular data on local energy and water supply capabilities, conducting a detailed analysis of land availability and planning risks, and assessing the long-term sustainability of each site. Engaging with local authorities and stakeholders will also be crucial to ensure alignment with regional development goals and community needs.

References

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Appendix A — Full Ranked Candidates Table

The table below shows the full set of evaluation scores for the top candidates. All scores are on a 0–10 scale. Water and climate scores are heuristic placeholders pending integration of dedicated datasets.

Region	Overall	Energy	Water	Climate	Latency	Resilienc e	Land	Plan Risk
Rugby	6.54	8.53	7.30	2.32	9.12	7.20	7.48	2.00
Welwyn Hatfield	6.45	7.70	7.30	1.74	6.87	7.20	9.93	2.00
Dacorum	6.42	7.87	7.23	1.74	7.32	7.20	9.35	2.00
Hertsmere	6.36	7.43	7.32	1.66	6.85	7.20	9.95	2.00
Newcastle-under-Lyme	6.39	8.06	7.29	2.91	8.55	7.20	6.90	2.00

This is a hackathon prototype using public open datasets and heuristic scoring. It is not an investment-grade site-selection tool. Scores marked as heuristic (water, climate) are placeholders until appropriate datasets are integrated. This report was generated automatically and has not been reviewed by a human expert.